Universiteit Antwerpen
Faculteit Toegepaste Ingenieurswetenschappen

5 – Infrastructuur & Beveiliging

Lab 2: Layer 2 switching and Layer 3 routing

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Table of Contents

1 PREPARATION 3

2 LAB GOALS 3

3 EXERCISE 1: SWITCHING WITH /16-SUBNET MASK 3

4 EXERCISE 2: SWITCHING WITH /24-SUBNET MASK 6

5 EXERCISE 3: SWITCHING WITH DIFFERENT SUBNETS FOR HOSTS 6

6 EXERCISE 4: ROUTING WITHOUT DEFAULT GATEWAY AND NO IP ROUTING 7

7 EXERCISE 5: ROUTING WITH DEFAULT GATEWAY AND NO IP ROUTING 8

8 EXERCISE 6: ROUTING WITH DEFAULT GATEWAY AND IP ROUTING 9

9 EXERCISE 7: ROUTING WITH DEFAULT GATEWAY, IP ROUTING, AND ROUTING PROTOCOL 10

REFERENCES ..... FOUT! BLADWIJZER NIET GEDEFINIEERD.

1 Preparation

Use the material below to prepare for the lab and for the short multiple-choice test at the start of the lab session. The goal is to **understand the concepts**, not to memorize. You will **not** be asked to reproduce commands, configuration syntax, or specific numbers such as timer values. Instead, you should be able to explain what happens in a network and why.

Start with the Packet Traveling series on Practical Networking. Read the articles on host-to-host communication directly, through a switch, and through a router [1]. Then read Role of Subnet Mask to understand how devices determine whether traffic can be sent directly or must be forwarded to a router [2].

Next, read Understanding Routing in Computer Networks to learn the difference between static and dynamic routing [3].

Finally, watch the short videos on RIP [4] and OSPF [5], and read this comparison of RIP and OSPF [6]. Focus on how each protocol shares routing information and how quickly it reacts when the network changes. You will see this difference for yourself during the lab.

2 Lab Goals

This lab introduces the fundamental concepts of Layer 2 switching and Layer 3 routing through a series of hands-on networking exercises. The goals of the lab today are the following:

- Configure and verify Layer 2 switching connectivity.
- Explain the effect of subnet masks and default gateways on network communication.
- Configure and verify Layer 3 routing between multiple subnets.
- Analyze ARP, ICMP, and routing traffic using Wireshark.
- Configure and validate static routes.
- Configure and compare dynamic routing using RIP and OSPF.

> **When your group finishes an exercise, make sure to alert the lecturer before continuing to the next exercise.**

3 Exercise 1: Switching with /16-Subnet mask

For the first exercise, the hardware setup of Figure 1 needs to be constructed. To make the configuration process easier, pair up with another student in your group. Each pair will be responsible for one switch. For the first labs that are not focused on security, turn off the firewall on each switch with the following command.

```
/etc/init.d/firewall stop
```

Start by configuring all hosts and switch management interfaces with the correct IP settings and `/16` mask. Do not configure any default gateways at this stage. Within each pair, **Student 1 will configure**

the IP address of the switch. Once the configuration is complete, physically connect the switches and hosts according to the topology in the figure. Make sure all devices are properly connected before continuing. It is important to verify physical connectivity on all links.

After your whole group has finished setting this up, fill in the questions. Try to help out your peers if they require any assistance.

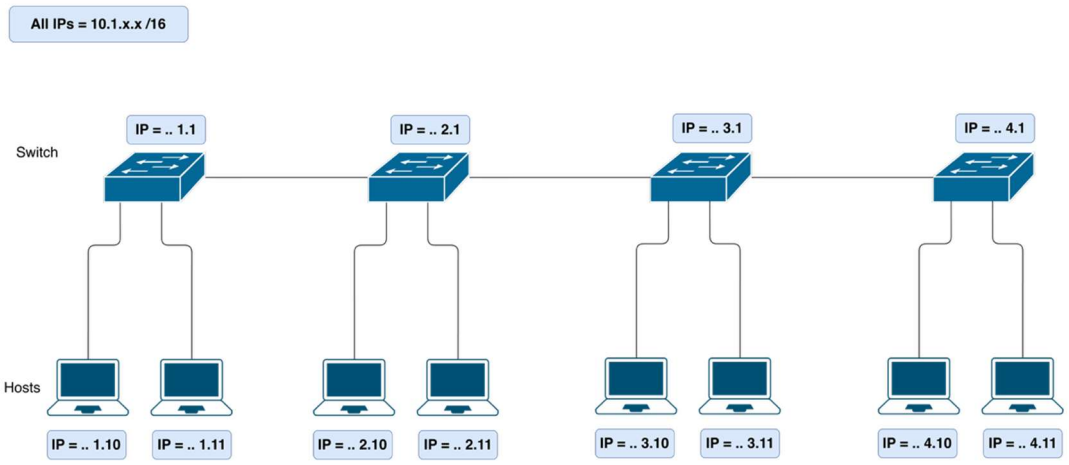


Figure 1: Setup for Layer 2 switching

Instead of pinging IP addresses one by one, you can use `fping` to check multiple hosts at once by listing them in a text file. This is a quick way to test connectivity across your network. Start by **creating a text file** (for example, `ips.txt`) in your Documents folder. Write all the IP addresses you want to ping, one per line. **Run** `fping` using the -f option to read the file:

```
fping -f ~/Documents/ips.txt
```

Ping your own switch management IP. Can you reach it?

ping 10.1.2.1

Ping another switch management IP. What do you get?

ping 10.1.1.1

Ping the IP of your partner. What is the result?

ping 10.1.2.10

Ping a few other hosts in your created network. Can you reach them?

ping 10.1.4.1

Let us now examine how this Layer 2 setup behaves internally. In this part of the lab, you will use **Wireshark** to observe the packets exchanged between the hosts and the switch. The goal is to understand how a simple **ping** moves through the network and how it can reach the other device. When a host wants to send an IP packet on an Ethernet network, it does not send it directly to an IP address. Ethernet forwarding happens using **MAC addresses**.

When a host starts a ping to another host in the same subnet, what information does it already know, and what information does it still need before it can send the Ethernet frame?

_____

_____

What is the very first relevant frame you observe in Wireshark when the ping starts? Note you might have to clear the table or ping a new host or switch.

_____

_____

Is that first frame addressed to a single device, or to multiple devices in the local network?

_____

_____

Paste a screenshot of the first packet you observe. Which addresses can you identify in that frame at the Ethernet level, and which addresses appear inside the protocol payload?

_____

_____

Paste a screenshot of the ICMP packet you observe. What can you find inside it?

_____

_____

Does the ICMP traffic use the same type of Ethernet destination addressing as the first frame you observed?

_____

_____

Which information seems to have been learned or cached after the first exchange?

_____

_____

In this experiment, which parts of the communication belong to Layer 2, and which belong to Layer 3?

_____

_____

Explain why no default gateway is needed in this topology?

Alle toestellen zitten in hetzelfde subnet (10.1.0.0/16). Elke bestemming is dus rechtstreeks bereikbaar via ARP en een Layer 2 switch. Een gateway is enkel nodig voor bestemmingen buiten het eigen subnet.

_____

What would you expect to break if the subnet mask were changed to `/24`?

_____

Enkel toestellen met hetzelfde derde octet blijven elkaar zien (dezelfde 10.1.x.0/24). Voor alle andere groepen valt de bestemming buiten het subnet, de host probeert de default gateway te gebruiken, maar die is er niet, dus er wordt niets verstuurd en de ping faalt.

4 Exercise 2: Switching with /24-Subnet mask

Use the same physical hardware topology that was built in the previous exercise, but now modify the network configuration so that all devices use a **`/24` subnet mask (`255.255.255.0`)** instead of the previous mask. The assigned **IP addresses must remain unchanged**, and **no default gateway may be configured** on any host or switch. For this exercise, **Student 2** is responsible for configuring the switch correctly before the group proceeds with the connectivity tests.

Ping your own switch management IP. Can you reach it?

ping 10.1.2.1

Ping another switch management IP. What do you get?

ping 10.1.2.1

Ping the IP of your partner. What is the result?

ping 10.1.2.10

Ping a few other hosts in your created network (other groups). Can you reach them?

ping 10.1.4.1

Connect physically to a different switch. Can you access the device via SSH? Why (not)?

Nee

Now open Wireshark on your devices. What do you see when you ping a host of another group? And why is this the case?

Niets. Geen ARP en geen ICMP. De host past het masker toe, ziet dat het doel buiten het eigen subnet ligt, vindt geen route (geen default gateway) en weigert lokaal te versturen. Er verlaat dus geen enkel pakket de netwerkkaart.

5 Exercise 3: Switching with different subnets for hosts

In this exercise, the goal is not to determine whether communication works, but to explain **why** a ping is or is not transmitted. Use the same physical topology as in the previous exercise, but configure the **hosts so that they are no longer all in the same IP subnet**. Do not configure a default gateway. Change the subnet of the host with the IP **`10.1.x.10`** to a **`/16`** subnet mask. The host with the **`10.1.x.11`** IP address will remain in the **`/24`** subnet. Switches can stay on the **`/24`** subnet. Analyze the network again.

Who can you reach, and why?

- Your partner on the same switch `10.1.x.10/.11`? YES, YES

- Your switch `10.1.x.1`? YES, YES
- Your neighbor on a different switch `10.1.x.10`? YES, NO
- Your neighbor on a different switch `10.1.x.11`? NO, YES
- Your neighbor's switch `10.1.x.1`? NO, NO

6 Exercise 4: Routing without default gateway and no IP routing

We will now extend the previous setup by adding routers on top. For this exercise, **Student 2** will configure the router. Start by installing the following pacakges on each router. You will need these later on in this lab session:

```
opkg update
opkg install frr frr-ripd frr-ospfd frr-zebra frr-watchfrr frr-vtysh
```

For convenience, disable the firewall on each router with the following commands.

```
/etc/init.d/firewall stop
```

Next, you will have to create a new interface with a static IP address and assign it to LAN2. First, remove LAN2 from the bridge device to isolate the networks. Then, create a new interface with the name **`lan2`**, and assign its IP address according to Figure 2. Once this is done, you can configure the IP address and subnet mask of the bridge interface (aka **`br-lan`**), which includes LAN1. No default gateway is needed for this exercise. Don't forget to save and apply your changes. Make sure all **subnet masks are `255.255.255.0 = /24`**, so that every duo is in another network.

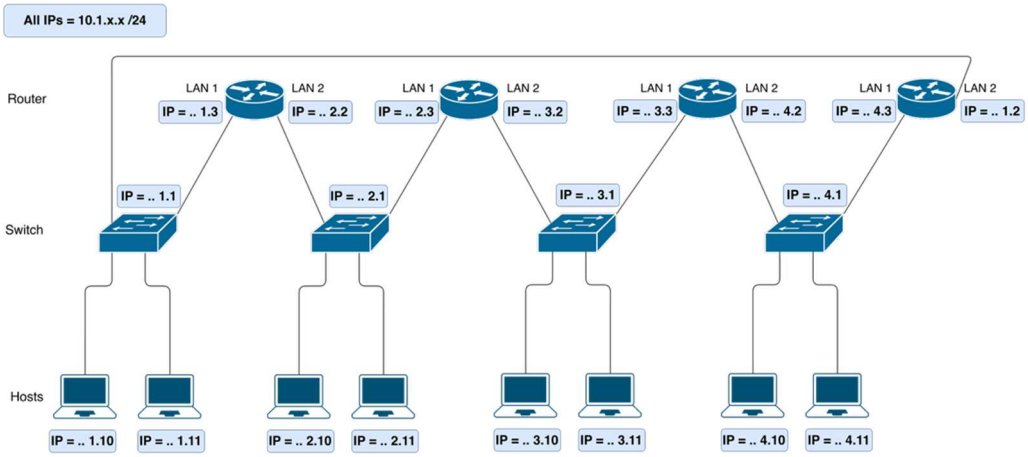


Figure 2: Setup for Layer 3 routing without default gateways

Who can you reach, and why?

- Your partner on the same switch `10.1.x..10/11`? _____
- Your switch `10.1.x.1`? _____
- Your router on the physically connected interface (LAN1) `10.1.x.3`? _____
- Your router on the other interface (LAN2) `10.1.x.2`? _____
- Your neighbor's router on the physically connected interface (LAN2) `10.1.x.2`? _____

- Your neighbor's switch? _____
- Your neighbor's router? _____

Open Wireshark and check what packets are sent and which are not. Why are some packets not sent?

_____

_____

Draw a conclusion: why don't you have full connectivity?

_____

_____

7 Exercise 5: Routing with default gateway and no IP routing

Configure the default gateways according to **Fout! Verwijzingsbron niet gevonden.**, i.e., the DGW of the `.10` host goes to the left side, the DGW of the `.11` host goes to the right side. The routers remain in unchanged.

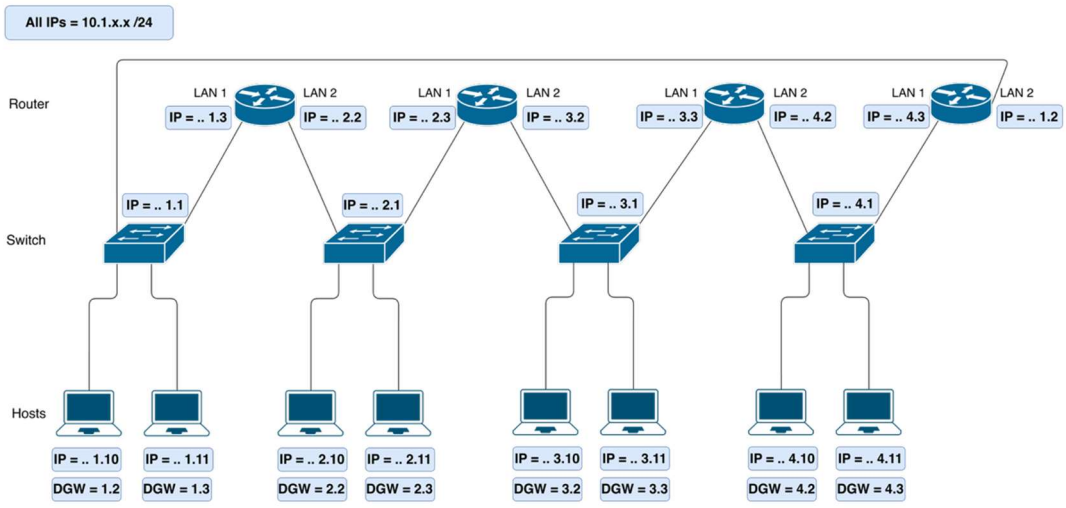


Figure 3: Setup for Layer 3 routing with a default gateway on the hosts.

Who can you reach, and why?

- Your partner on the same switch `10.1.x.10/.11`? _____
- Your switch `10.1.x.1`? _____
- Your router on the physically connected interface (LAN1) `10.1.x.3`? _____
- Your router on the other interface (LAN2) `10.1.x.2`? _____
- Your closest neighbor `.10/.11`? _____
- Your closest neighbor's switch? _____
- Your farther neighbor's switch? _____
- Your farther neighbor's router? _____

Open Wireshark on your host device and send a ping to the far side of the router you can reach. What do you learn from this?

_____

_____

Draw a conclusion: why don't you have full connectivity?

_____

_____

8 Exercise 6: Routing with default gateway and IP routing

Keep the same physical topology and host default gateways. In the previous exercise, the hosts knew where to send packets for another subnet because a default gateway was configured, but the routers only knew their directly connected networks. In this exercise, you will extend the router configuration by adding **static routes** so that packets can also be forwarded to more distant subnets. A static route is simply a manually configured destination prefix together with the gateway to which packets for that prefix must be sent. In OpenWrt, static IPv4 routes can be configured on a specific interface, and routing itself is handled by the kernel routing table.

On each router, add the required static route(s) such that you get full reachability in the network. Agree with your group on how and where to add these static routes.

> **Tip: Each router is directly connected to two subnets, so you will have to add two static routes per router to reach the other two subnets.**

How did you add the static route, and on which interface? What IP address will be the target, and what will be the gateway?

_____

_____

With what command can you check the routing table? Explain what you see

_____

_____

Who can you reach, and why?

- Your partner on the same switch `10.1.x.10/.11`? _____
- Your switch `10.1.x.1`? _____
- Your router on the physically connected interface (LAN1) `10.1.x.3`? _____
- Your router on the other interface (LAN2) `10.1.x.2`? _____
- Your closest neighbor `.10/.11`? _____
- Your closest neighbor's switch? _____
- Your farther neighbor's switch? _____
- Your farther neighbor's router? _____

What would be a disadvantage of static routes for real networks?

_____

_____

Let's say we have full connectivity by adding all static routes. If a link or router fails, what happens to the static routes that use that path? Will the network adapt automatically? Why or why not?

_____

_____

Open Wireshark on your host and ping a host in a remote subnet. Which MAC address does your host use as destination in the Ethernet frame: the MAC address of the final host, or the MAC address of the default gateway? Why?

_____

_____

Inspect the IP packet in Wireshark on both sides of the router. Which fields stay the same, and which field(s) may change when the router forwards the packet?

_____

_____

9 Exercise 7: Routing with default gateway, IP routing, and routing protocol

In the previous exercise, you manually added static routes to get connectivity. This quickly becomes unmanageable as the network grows; every topology change means manually updating every router. In this exercise, you will replace those static routes with dynamic routing protocols. You will configure the same topology two times, each time with a different protocol: **RIP** and **OSPF**. By the end, you should understand why different protocols exist and when you would choose one over another. To run these routing protocols, you will use **FRR (Free Range Routing)**, an open-source routing software suite that implements multiple dynamic routing protocols on the router.

Before continuing, **remove all static routes** you added in the previous exercise via `Network → Routing`. Normally, you should have installed the FRR packages at the start of "Exercise 4: Routing without default gateway and no IP routing".

9.1 RIP (Routing Information Protocol)

RIP is a distance-vector protocol, the oldest dynamic routing protocol still in common use. Each router periodically sends its entire routing table to its neighbors. Routers learn about remote networks hop by hop, and the metric is simply hop count. RIP has a maximum of 15 hops, making it unsuitable for large networks, but it is very easy to understand and configure, which makes it a great starting point.

Step 1: Enable the RIP daemon on every router

```
vi /etc/frr/daemons
```

Set `ripd=yes`, then restart FRR:

```
/etc/init.d/frr enable
/etc/init.d/frr restart
```

Step 2: Configure RIP on every router

For `Router x` (adapt network statements to your own IP ranges):

```
vtysh
conf t
router rip
 version 2
 network 10.1.x.0/24
 network 10.1.(x+1).0/24
 no passive-interface br-lan
 no passive-interface lan2
end
write memory
exit
```

Each router should advertise both of its directly connected networks. To verify the setup, you can move between the `vtysh` and the regular OpenWrt command line by using the command

```
vtysh
```

and

```
exit
```

What commands can you find to debug this setup and analyze the routing table?

_____

_____

Explain the output of the commands. What metric does it show for a network two hops away? What does this number represent?

_____

_____

Ping all hosts, switches, and routers. Until where do you have connectivity?

_____

_____

Determine the route the packets travel using traceroute.

_____

_____

What route is taken to the closest neighbors?

_____

_____

What route is taken to the neighbors of your neighbors?

_____

_____

Are these always the best possible routes?

_____

_____

Now, disconnect a link between a switch and a router. What happens? How long does it take ("bad news travels slow/fast?")

_____

_____

Reconnect the link. What happens? ("good news travels slow/fast?")

_____

_____

Open Wireshark on the inter-router link. What destination IP address do RIP update packets use, and why is this significant?

_____

_____

What is the convergence behavior if you unplug the cable between two routers? How long until RIP detects the failure and updates its routing table? This is called the convergence time.

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Open Wireshark and sniff between the routers. Can you find out what is being transmitted? Capture it and dissect the packet!

_____

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9.2 OSPF (Open Shortest Path First)

OSPF is a link-state protocol. Instead of sharing routing tables like RIP, each router shares information about its directly connected links with every other router in the network. Every router builds a complete map of the network (called the LSDB — Link State Database) and independently calculates the shortest path using Dijkstra's algorithm. OSPF converges much faster than RIP and scales to large networks.

Step 1: Disable RIP and enable OSPF on every router

Set `ripd=no` and `ospfd=yes`, then restart FRR.

```
vi /etc/frr/daemons
```

Step 2: Configure OSPF via `vtysh` on every router.

For `Router x`:

```
vtysh
conf t
router ospf
 ospf router-id 10.1.x.3
 network 10.1.x.0/24 area 0
 network 10.1.(x+1).0/24 area 0
end
write memory
```

Every router must have a unique `router-id`. Use its LAN1 IP address for clarity. All networks go into `area 0` (the backbone area) for this lab.

Ping all hosts, switches, and routers. Until where do you have connectivity?

_____

_____

Determine the route the packets travel using traceroute.

_____

_____

What route is taken to the closest neighbors?

_____

_____

What route is taken to the neighbors of your neighbors?

_____

_____

Are these always the best possible routes?

_____

_____

Now, disconnect a link between a switch and a router. What happens? How long does it take ("bad news travels slow/fast?")

_____

_____

Reconnect the link. What happens? (“good news travels slow/fast?”)

_____

_____

Run `show ip ospf neighbor`: what states do you see? What does the Full state mean?

_____

_____

Run `show ip ospf database`: what types of LSAs (Link State Advertisements) can you identify? What information does each one carry?

_____

_____

Compare convergence time to RIP: unplug the same cable as before. How quickly does OSPF detect the failure and reroute?

_____

_____

Look at the routes installed by OSPF with `show ip route`. How does OSPF decide which path is best when there are multiple paths to the same destination?

_____

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What is the fundamental difference in how RIP and OSPF routers share information with each other?

_____

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